SymPy Bug Card

Bug 61: Spurious Imaginary Term in Positive-Branch `expint` Series

Task. Expand the positive-real exponential integral at the origin.

$$\mathrm{series}(\mathrm{expint}(2, x), x, 0, 2), \qquad x > 0.$$

SymPy's answer.

$$1 + x(\log x - 1 + \gamma - i\pi) + O(x^2).$$

Correct answer.

$$1 + x(\log x - 1 + \gamma) + O(x^2).$$

Explanation. For $x > 0$, $E_2(x) = e^{-x} - xE_1(x)$ and $E_1(x) = -\gamma - \log x + x + O(x^2)$, so the first-order expansion is real. The returned $-i\pi x$ term makes the truncated series complex at positive real inputs, for example at $x = 10^{-3}$.

Likely root cause. The diagnosis traces the call through `expint._eval_nseries`, which rewrites to $\mathrm{Ei}(x\ \mathrm{exp\_polar}(i\pi))$; then `Ei._eval_rewrite_as_Si` treats that polar argument as an ordinary negative argument and subtracts $i\pi$. This is a new instance of a known failure mode.

Artifact (GitHub). `bug-report/bug-061-spurious-imaginary-term-in-positive-branch-expint-series/`